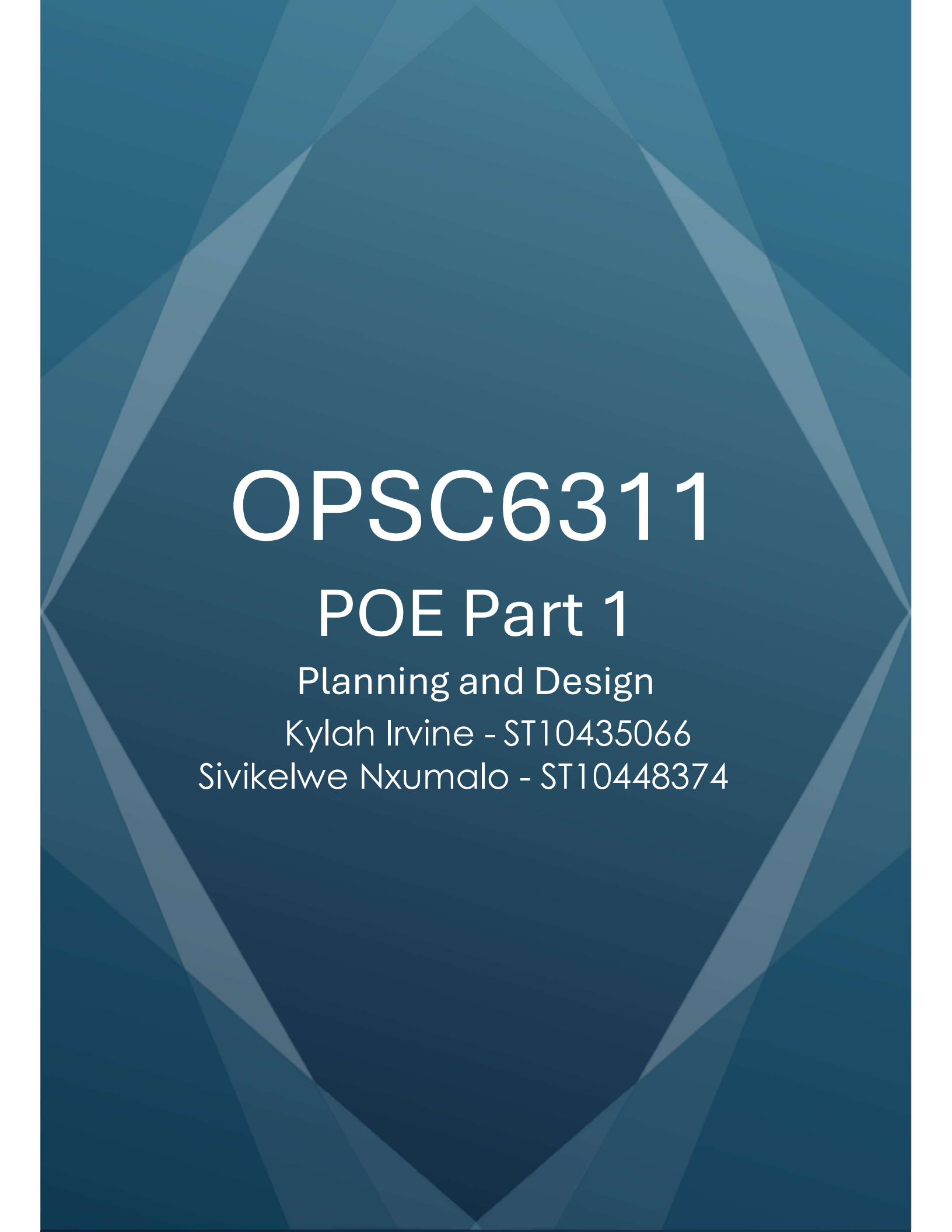




OPSC6311

POE Part 1

Planning and Design

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Table of Contents

Introduction	3
Planning	4
Functional Requirements.....	4
Non-Functional Requirements	5
Design	6
Visual Identity & Colour Scheme.....	7
Colour Scheme:	7
Mock-up Descriptions & Logic	8
Screen 1: Secure Entry (Login)	8
Screen 2: Home Dashboard	9
Screen 3: The Split	9
User Scenario: A Day with Tia.....	10
User Journey (Human Computer Interaction Focus)	11
Glossary (Appendix)	12
Gantt Chart	13
Reference List.....	15

Table of Figures

Figure 15 Official Centric Logo.....	6
Figure 16 Centric Application Icon	6
Figure 17 Centric Log in and create account.....	8
Figure 18 Centric Home Dashboard	9
Figure 19 Centric Screen Split	9
Figure 20 Gantt Chart	13

Introduction

Moving toward a cashless society has made it intimidatingly tough for people, particularly students, to keep track of where their money goes. Between quick tap-to-pay meals and endless subscriptions, it is easy to lose sight of a monthly budget before the first week is even over.

This report looks into three major players in the budget app market (Mint, YNAB and Buddy) to figure out precisely what kind of design and logic is currently needed to build a better budgeting tool for the Android platform. The goal this phase of the development process (part 1) is to take the best features of each application we have conducted research on and form an initial design. The end goal of the POE is to create an Android application, including the following features: the automation found in Mint, the strict discipline of YNAB and the social features of Buddy, of our own that specifically tailors to the South African student experience.

Planning

Functional Requirements

User Registration & Login

- The user must be able to register a new account and log in to the app using a username and password that the user set themselves during registration.

User Secure Onboarding

- The app must allow users to create a local profile and set a password for encrypted entry.

Expense Categories

- The user must be able to create categories for expenses so they can sort their transactions.

Expense Entry

- The user must be able to add a transaction and have that transaction saved.
- The user must include the amount, the date and some form of description when adding the transaction. This transaction should be added to the desired category.
- The user must be able to optionally attach a photo to each expense entry like say a receipt or bill.
- Users must have the ability to create new entries, read their spending history, update categories and delete incorrect transactions (CRUD).

Budget Goals

- The user must be able to set a monthly total budget goal and specific limits for individual categories.

Expense Viewing

- The user must be able to view a list of all expenses during a user-selectable period.
- If a receipt photo was stored for an entry, the user must be able to access it from this list.
- The user must be able to view the total amount spent on each category during a user-selectable period.

Automated Split Engine

- Based on our research into the Buddy app, Centric must allow a user to toggle a "Split" on any transaction. The system must automatically calculate the user's portion based on the custom percentage slider, which allows the user to determine their ratio of the split.

Data Storage

- All this data should be saved to a local DB.

Graphs & Dashboards (Final POE only)

- The user must be able to view a graph showing daily spending over a user-selectable period. This graph must clearly display category spending and allow the user to identify trends.
- The app must display a progress dashboard that shows how well the user is staying within their budget goals for the current month. Overspending categories must be highlighted visually.

Gamification (Final POE only)

- The app must include gamification elements such as rewards or badges for meeting budget goals or consistent expense logging.

Real-Time Budget Visualisation:

- The system must feature a dynamic "Remaining Allowance" ring that updates instantly as data is entered.

Non-Functional Requirements

Responsiveness & Speed

- The UI must respond to touch inputs fast to keep the Neo-Minimalist feel.

Privacy

- All data must be stored locally on the device (not given to the company) to make the app more "private" and mitigate the risk of big data leaks that jeopardise user data integrity.

Reliability

- The app run at least 99.9% of the time without crashing or any other catastrophic failures.

Accessibility (WCAG Compliance)

- The colour scheme (Teal and Coral) must maintain high contrast ratios (lightest teal was changed in pre-design to be more contrasted) to ensure the app is usable for individuals with visual impairments and to fit different colour themes.

Design

Centric is a personal budgeting application designed to bridge the gap between complex professional financial tools and the collaborative needs of the modern South African user. While the app is built to be accessible to everyone, it specifically caters to the student demographic by addressing challenges such as shared living expenses and "invisible" digital spending.



The Centric logo was custom designed using Canva, focusing on a minimalist aesthetic that reflects the app's core value of financial unity. The interlocking "C" and bar graph elements were specifically chosen to represent growth and integrated tracking.

Figure 1 Official Centric Logo

The application icon, also developed within Canva, serves as the primary visual touchpoint for the Android interface. It conforms to modern mobile design standards by using a rounded square frame and the established brand colour palette to ensure high visibility and brand recognition on the user's home screen.



Figure 2 Centric Application Icon

Visual Identity & Colour Scheme

The design follows the 2026 Neo-Minimalism trend, which prioritizes a "Simplicity with Soul" approach. Using clean layouts paired with soft shadows and gradients to build user trust.

Colour Scheme:

Primary Teal (#26A69A): Indicates positive financial growth and "Safe-to-Spend" zones.

Midnight Navy (#1A237E): Used for primary headers and navigation bars to provide a sense of stability.

Coral Alert (#FF7043): An accent colour for critical warnings, overspending and split notifications.

Glassy Neutral: Semi-transparent "Glassmorphism" layers for card backgrounds to create a sense of depth and modernity.

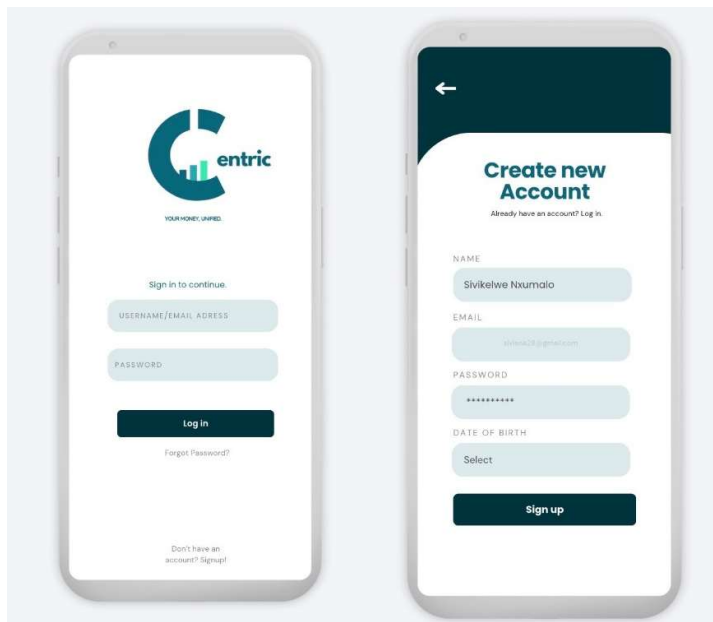
Mock-up Descriptions & Logic

All visual assets and interface screens for Centric were custom designed using Canva. The design logic applies Neo-Minimalism and Glassmorphism, securing the application feels like a modern, premium tool rather than a stressful financial spreadsheet.

Screen 1: Secure Entry (Login)

Detailed Description

This screen features a semi-transparent, frosted-glass login module ("Glassmorphism") centred over a soft Navy-to-Black gradient background. The input fields for Username and Password use rounded corners to match the Neo-Minimalist aesthetic.



Interaction Logic

To reduce the cognitive load, the screen is stripped of all non-essential elements. When the user enters their password, a Haptic feedback response is triggered on the device (a vibration, that can be turned off in the application settings for users with sensory preferences). This provides a tactical sense of security and confirmation.

Figure 3 Centric Log in and create account

Screen 2: Home Dashboard

Detailed Description

The dashboard is dominated by a large, high contrast and circular “money remaining” widget. Beneath it is a section that displays individual transaction history items, presented as floating cards.

Interaction Logic

A floating action button with an addition sign “+” is placed on the bottom- right corner for one-handed thumb access.

The “Remaining Allowance” ring acts as a live indicator of. It remains Teal until the user exceeds 80% of their budget. If the user exceeds 80% of the set budget, the ring will shift to Coral alert (shown by the red-orange rectangle in the corner displaying that the user has R170.00 to spend before they reach the 80% mark and the ring changes).

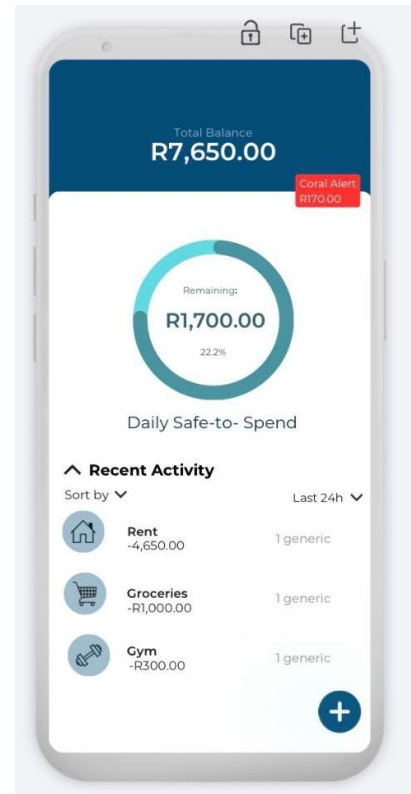


Figure 4 Centric Home Dashboard

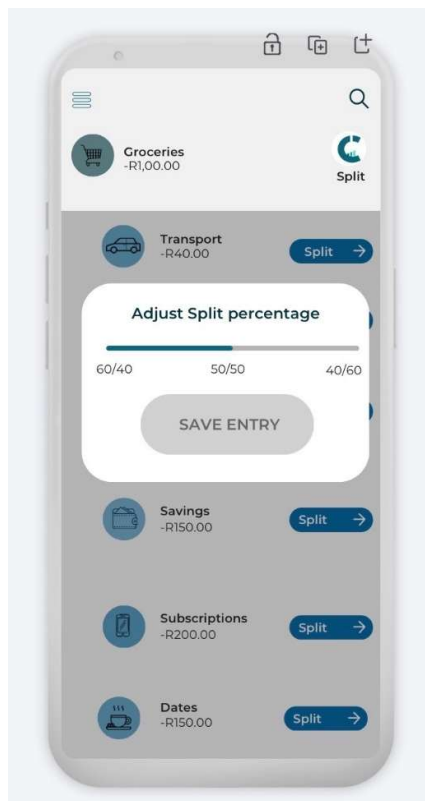


Figure 5 Centric Screen Split

Screen 3: The Split

Detailed Description

This interface provides a clean, vertically scrollable list of recent expenses. Each item includes a prominent toggle switch labelled "Split."

Interaction Logic

This feature is tailored for shared student living (for example: roommates). Activating the toggle reveals a sliding percentage bar (defaulting to 50/50). When saved the app executes a background calculation: it updates the user's personal budget with their share and automatically moves the balance to an "Owed to Me" ledger.

User Scenario: A Day with Tia

Character Profile

- Name: Tia
- Status: Final-year Software Development Student
- Goal: Manage her weekly allowance while splitting res-costs with her roommate.

The Morning Login Tia wakes up and needs to check if she has enough data money for her online lecture. She opens Centric. She enters her password and the soft haptic vibrations give her a tactile confirmation that she's logged in securely.

The Dashboard Insight Tia lands on her Home Dashboard. She looks at the big Teal ring and sees she has R350 left for the week. She scrolls through her floating cards and notices her grocery spend from yesterday. Seeing the ring is still Teal gives her immediate peace of mind. She does not have to do any math to know she is still on track. She taps the "+" Floating Action Button and quickly logs a R15 bread purchase before heading out.

Managing the Split feature Later, Tia's roommate pays the R400 electricity bill for their room. Tia opens The Split screen, finds the electricity transaction, and presses the "Split" button. She slides the bar to 50/50. Instantly, the app tells her that her share is R200. She hits confirm, and her dashboard updates her "Remaining Money" without her ever having to open a calculator. The app has done the heavy lifting, allowing Tia to focus on her studies instead of her stress.

User Journey (Human Computer Interaction Focus)

To ensure the app is intuitive, the following "User Stories" demonstrate the app's ease of use:

- The Quick Log: "As a user, I want to tap the '+' button and enter 'R40' for a coffee in under 5 seconds so that I don't forget to track my spending."
- The Roommate Split: "As a student, I want to split a R200 electricity bill with my roommate so that my personal budget only reflects my actual R100 expense."
- The Goal Check: "As a user, I want to see a green progress bar for my 'New Laptop' savings goal so I stay motivated to save."

Glossary (Appendix)

CRUD: Create, Read, Update, and Delete (the four actions the app performs on data).

HCI (Human-Computer Interaction): The science of making apps easy and comfortable for humans to use.

Neo-Minimalism: A modern design style that uses simplicity, soft shadows, and gradients.

Room Database: A secure "storage locker" on your phone for your financial data.

Gantt Chart

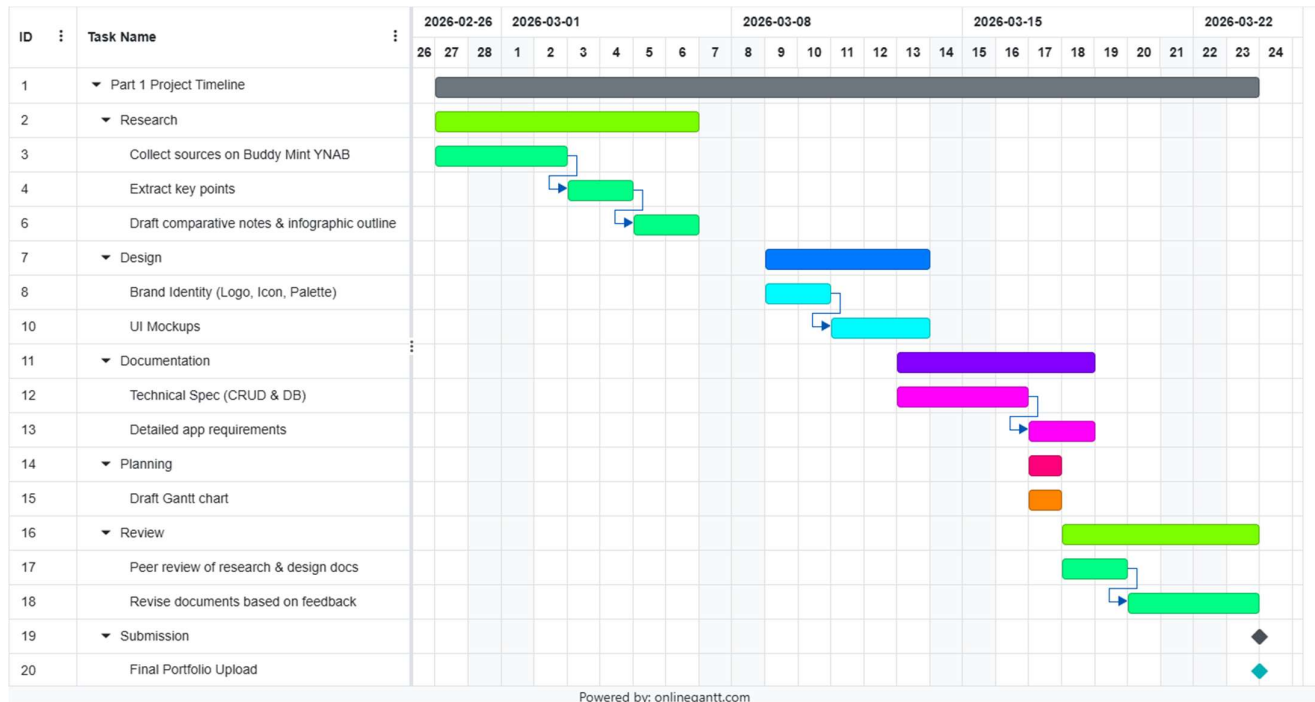


Figure 6 Gantt Chart

We created the Gantt Chart and project timeline to represent the timeline for Part 1 (research, planning and design phases). We cannot plan for the final submission as of yet due to not being provided the due date of part 3.

Phase:	Task Description:	Start Date:	End Date:	Duration:
Research	Analysis of Mint, YNAB, and Buddy frameworks.	Feb 27	Mar 05	7 Days
Design	Brand Identity: Logo, Icon, and Palette creation.	Mar 06	Mar 07	2 Days
Design	UX Architecture: High-fidelity UI mockups.	Mar 10	Mar 12	3 Days
Documentation	Technical Spec: CRUD & Database requirements.	Mar 13	Mar 15	3 Days
Planning	Gantt Chart: Finalizing project timeline.	Mar 17	Mar 17	1 Day
Review	Quality Control: Peer review and revisions.	Mar 20	Mar 22	3 Days

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Submission	Final Portfolio Upload.	Mar 24	Mar 24	1 Day
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Reference List

Canva Design Lab, 2025. Brand Identity Systems for Mobile Startups. [Online] Available at: <https://www.canva.com/design> [Accessed 19 March 2026].

Canva Design Lab, 2025. Brand Identity Systems for Mobile Startups. [Online] Available at: <https://www.canva.com/docs/brand-guidelines/> [Accessed 23 March 2026].

Cape Town FinTech Review, 2026. Digital Wallet Fatigue: Why Students are Returning to Manual Tracking. South African Finance Journal, 12(2), pp. 45-50.

Cloud Fusion, 2026. Mastering Mobile Application Design & Development in 2026. [Online] Available at: <https://www.cloudfusion.co.za/blog/master-mobile-application-design-development-2026-en> [Accessed 22 March 2026].

Dark Square, 2026. UI Design Trend 2026 #1: Neo-Minimalism. [Online] Available at: <https://www.darksquare.com/trends/2026/neo-minimalism> [Accessed 23 March 2026].

Dark Square, 2026. UI Design Trend 2026 #1: Neo-Minimalism: Simplicity With Soul. [Online] Available at: <https://www.darksquare.com/trends/2026/neo-minimalism> [Accessed 20 March 2026].

Financial IT, 2026. Buddy – The Financial Companion Making Budgeting Fun and Easy. [Online] Available at: <https://financialit.net/news/personal-finance/buddy-financial-companion-making-budgeting-fun-and-easy> [Accessed 23 March 2026].

Forbes Advisor, 2026. Best Budgeting Apps of 2026. [Online] Available at: <https://www.forbes.com/advisor/banking/best-budgeting-apps> [Accessed 23 March 2026].

Intuit, 2026. The Evolution of Mint into Credit Karma. [Online] Available at: <https://www.creditkarma.com/about/releases/intuit-credit-karma-welcomes-all-minters> [Accessed 23 March 2026].

Investopedia, 2026. YNAB vs. Mint: Which Is the Better Budgeting App? [Online] Available at: <https://www.investopedia.com/ynab-vs-mint-5179966> [Accessed 23 March 2026].

Koseoglu, S., 2026. The New Language of Quiet Power: Neo-Minimalism in UI Design. Journal of Digital Aesthetics, 8(1), pp. 12-25.

Level Access, 2026. Digital Accessibility Standards: WCAG 2.1 Level AA for Mobile Apps. [Online] Available at: <https://www.levelaccess.com/accessibility-2026> [Accessed 20 March 2026].

Level Access, 2026. Digital Accessibility Standards: WCAG 2.1 Level AA for Mobile Apps. [Online] Available at: <https://www.levelaccess.com/blog/wcag-for-mobile-apps/> [Accessed 23 March 2026].

Mintos Blog, 2025. Life After Mint: Finding the Right Tracker for Analysis of Budgeting Apps. [Online] Available at: <https://beancount.io/forum/t/life-after-mint-finding-the-right-alternative-for-your-financial-style/99> [Accessed 23 March 2026].

Mobile Design Monthly, 2026. Top UI/UX Mobile App Design Trends To Watch In 2026. [Online] Available at: <https://www.mobiledesignmonthly.com/trends2026> [Accessed 21 March 2026].

Moneywise, 2026. Mint.com Review | A Budgeting and Tracking App. [Online] Available at: <https://moneywise.com/managing-money/budgeting/mint-com> [Accessed 23 March 2026].

Mustachian Post, 2026. YNAB Review 2026: The Best App to Build a Budget. [Online] Available at: <https://www.mustachianpost.com/ynab-review> [Accessed 23 March 2026].

NerdWallet, 2025. YNAB App Review for 2025. [Online] Available at: <https://www.nerdwallet.com/finance/learn/ynab-app-review> [Accessed 23 March 2026].

NerdWallet, 2026. Why YNAB is Still the King of Zero-Based Budgeting. [Online] Available at: <https://www.nerdwallet.com/article/finance/ynab-app-review> [Accessed 23 March 2026].

Nielsen Norman Group, 2026. 10 Usability Heuristics for Financial Mobile Interfaces. [Online] Available at: <https://www.nngroup.com/articles/ten-usability-heuristics/> [Accessed 23 March 2026].

Nielsen, J., 2026. 10 Usability Heuristics for Financial Mobile Interfaces. [Online] Nielsen Norman Group. Available at: <https://www.nngroup.com/articles/ten-heuristics-2026> [Accessed 21 March 2026].

PCMag, 2026. The Best Personal Finance and Budgeting Apps We've Tested. [Online] Available at: <https://www.pcmag.com/picks/the-best-personal-finance-services> [Accessed 23 March 2026].

Policygenius, 2026. The Most Popular Free Budgeting Apps, Reviewed. [Online] Available at: <https://www.policygenius.com/personal-finance/news/a-look-at-the-five-most-popular-budgeting-apps> [Accessed 23 March 2026].